Qualivio

PHARMACOVIGILANCE · QUALITY ASSURANCE | LIFE SCIENCES

BEGINNER'S GUIDE

Pharmacovigilance Made Simple

A clear, plain-English introduction to how medicines are monitored for safety after they reach patients, told through the story of one man and one new drug.

Beginner Friendly

Free Resource

2026 Edition

CONTENTS

What's Inside

This guide uses the story of Martin, a real-life inspired patient, to walk you through the entire pharmacovigilance process from a suspected side effect to a label update. No prior knowledge required.

📚

HOW TO USE THIS GUIDE

Read it from start to finish to follow Martin's story, or jump straight to any section that interests you. Each chapter stands on its own.

CHAPTER 01

What Is Pharmacovigilance?

Before a medicine reaches patients, it goes through years of testing. But no clinical trial can predict every risk for every person. That is where pharmacovigilance comes in.

DEFINITION

Pharmacovigilance is the science and activities relating to the detection, assessment, understanding, and prevention of adverse effects or any other drug-related problems.

In plain terms: it is the ongoing process of watching out for problems with medicines once they are being used in the real world, and acting on what is found.

Why Clinical Trials Are Not Enough

Clinical trials are carefully controlled. They involve thousands of volunteers over a limited time. This is enough to detect common side effects, but not rare ones. A side effect that occurs in one in every 10,000 patients might never appear in a trial of 3,000 people.

Once a medicine is approved and millions of patients take it, rare risks can begin to show up. Pharmacovigilance is the system that catches them.

130+
Countries in the WHO drug monitoring programme

30M+
Adverse event reports in VigiBase (global database)

1 in 10K
How rare a side effect can be and still get detected

Who Is Involved?

Healthcare professionals

Doctors, pharmacists, and nurses report suspected side effects they observe in patients.

Pharmaceutical companies

Manufacturers are legally required to monitor, collect, and report safety information.

Regulatory authorities

Agencies like the EMA or FDA review reports and take action when a risk is confirmed.

Patients

Patients can also report side effects directly in many countries, adding a vital perspective.

CHAPTER 02

Meet Martin

Martin, 61

Retired teacher · Lives with osteoarthritis

Martin is 61 years old and has osteoarthritis in his knees. His pain has gradually worsened, and it is no longer well controlled with exercise and occasional non-prescription pain relief.

His doctor decides it is time to try a prescription medicine. Martin is prescribed **Relvian**, a new anti-inflammatory drug that has recently been approved and launched on the market.

Starting the New Medicine

Martin collects his prescription from the local pharmacy. The pharmacist explains the known potential side effects: indigestion, dizziness, and mild headache. These are the risks identified during clinical trials.

Martin starts taking Relvian as directed and hopes his knee pain will soon improve.

GOOD TO KNOW

When a medicine is first approved, the product information lists only the side effects found during clinical trials. Side effects that are very rare may not appear until the drug has been used by a much larger population.

Something Unexpected Happens

After about one week, Martin notices new symptoms that were not on the pharmacist's list:

- Unusual shortness of breath when climbing stairs
- A persistent dry cough
- More fatigue than usual

Martin wonders whether Relvian could be responsible. He makes an appointment with his doctor.

PATIENT TIP

If you start a new medicine and notice any new or unusual symptoms, always tell your doctor or pharmacist, even if you are not sure whether the medicine is the cause. Your report could help protect other patients.

CHAPTER 03

A New Symptom Appears

Martin’s doctor takes his new symptoms seriously. Investigating a potential drug side effect requires careful and systematic work.

The Doctor’s Investigation

Martin’s doctor examines him thoroughly and orders a set of tests:

- Chest imaging (X-ray and CT scan)
- Blood tests to check for infection or inflammation
- Review of all other medications to rule out alternative causes
- Additional tests to exclude underlying lung disease

📋 CLINICAL FINDING

The results show signs of **pneumonitis caused by the medicine**: inflammation of the lung tissue most likely triggered by Relvian. No other cause is found. The doctor concludes Relvian is the probable cause.

Immediate Action

The doctor advises Martin to stop taking Relvian immediately. Over the following two weeks, Martin’s breathing gradually improves. The cough disappears. His oxygen levels return to normal.

💡 What is pneumonitis caused by a medicine?

Pneumonitis is inflammation of the lung tissue. When caused by a medicine, it is referred to as medicine-caused pneumonitis. Symptoms can include shortness of breath, dry cough, and fatigue, very similar to what Martin experienced. It is a rare but serious adverse reaction that is now documented for Relvian.

Causality Assessment

One of the key tasks in pharmacovigilance is determining whether a medicine actually caused a suspected adverse event. Doctors and regulators use established criteria, looking at:

📅 Timing

Did the symptoms appear after the medicine was started? Did they improve when it was stopped?

🔍 Alternative causes

Could the symptoms be explained by an infection, an existing condition, or another drug?

📊 Known data

Are there other reports of the same medicine causing the same type of reaction?

🔄 Rechallenge

If the patient takes the medicine again, do the same symptoms return? (This is rarely done intentionally.)

CHAPTER 04

Reporting the Adverse Event

Martin's doctor believes Relvian may have caused a side effect that is not yet listed in the product information. He has both a professional responsibility and a moral duty to report it.

The Individual Case Safety Report (ICSR)

Martin's doctor completes what is known as an **Individual Case Safety Report** (ICSR). This is the standard document used to report a suspected adverse event to a regulatory authority.

> **WHAT AN ICSR CONTAINS**
>
> A valid ICSR must include: (1) an identifiable patient, (2) an identifiable reporter, (3) the name of the suspected medicine, and (4) a description of the suspected adverse event. Martin's case provides all four.

Serious vs Non-Serious Adverse Events

Regulatory authorities classify adverse events by severity. Martin's case is considered **serious** because his condition required clinical investigation and medical intervention.

Serious

- Life-threatening
- Requires hospitalisation
- Results in disability
- Medically significant

Non-Serious

- Mild indigestion
- Temporary dizziness
- Mild headache
- Self-resolving symptoms

Where Does the Report Go?

1. **Doctor completes the ICSR**
   Martin's doctor records the case details and submits the report to the national regulatory authority.
2. **National authority receives the report**
   The report enters the national pharmacovigilance database where it is reviewed and processed.

3

Pharmaceutical company is notified

The manufacturer of Relvian is also required to receive and process this report, adding it to their own safety database.

CHAPTER 05

Into the Safety Database

One report on its own tells us about Martin's experience. Many reports together begin to reveal patterns. That is why safety databases exist.

The National Database

Martin's case is entered into the national adverse event database. Every country that monitors medicines maintains such a database. It collects reports from doctors, pharmacists, nurses, and patients, building up a picture of each medicine's real world safety profile over time.

🌏 GOING GLOBAL

Because Martin's country is a member of the **WHO Programme for International Drug Monitoring**, his report is shared with **VigiBase**, the WHO's global database of adverse drug reaction reports. VigiBase holds over 30 million reports from more than 130 countries.

How Reports Are Processed

1. **Data entry and coding**
   The adverse event is coded using standardised medical terminology (MedDRA), so that reports from around the world can be compared consistently.
2. **Quality check**
   The report is reviewed for completeness. Follow-up may be requested from the reporting doctor to gather more clinical detail.
3. **Duplicate check**
   Steps are taken to ensure the same event has not already been reported, which would lead to double-counting in the database.
4. **Sharing with VigiBase**
   Qualified reports are transmitted to VigiBase, where they join millions of other cases from around the world.

DID YOU KNOW?

The pharmaceutical company that manufactures Relvian also has its own safety database. By law, they must collect, process, and report adverse events to regulatory authorities. They are also required to periodically submit detailed safety analyses called **Periodic Benefit-Risk Evaluation Reports** (PBRERs).

CHAPTER 06

Detecting a Signal

One case can raise suspicion. A cluster of similar cases can generate a *signal*: the first evidence that a medicine may be linked to a previously unrecognised risk.

DEFINITION: SAFETY SIGNAL

A **signal** is information from one or more sources suggesting a new, potentially causal association between a medicine and an adverse event, or a new aspect of a known association, that warrants further investigation.

What Happens at the Regulatory Authority

A case assessor at the national regulatory authority reviews Martin's report. In doing so, they search both the national database and VigiBase and find a small number of similar reports: other patients taking Relvian who also developed lung inflammation.

This cluster of cases is evaluated more closely. Statistical methods are used to assess whether the number of reports is higher than would be expected by chance. Further clinical analysis suggests it is probable that Relvian is associated with lung inflammation caused by the medicine.

Signal Evaluation Steps

STEP 1: DETECTION
A pattern is noticed: multiple reports of lung inflammation in Relvian users, from different countries and different reporters.

STEP 2: VALIDATION
The reports are reviewed to confirm they are genuine, not duplicates, and that they describe the same type of event.

STEP 3: ANALYSIS
Clinical evidence is examined. Statistical tools help determine whether the association is greater than expected by chance.

STEP 4: PRIORITISATION
The signal is assessed for its severity, public health impact, and urgency. Drug-induced pneumonitis is serious, so it is prioritised.

STEP 5: RECOMMENDATION
Regulators conclude that action is warranted. The signal becomes a confirmed safety finding requiring a regulatory response.

CHAPTER 07

Regulatory Action and Label Update

When a confirmed safety signal points to a real risk, regulators act. Their goal is to protect future patients while keeping beneficial medicines available.

Updating the Product Information

The regulatory authority decides that Relvian's product information must be updated. The manufacturer is instructed to make two key changes:

📝

LABEL CHANGES FOR RELVIAN

Addition to adverse effects list: "Inflammation of the lungs (pneumonitis)" is added as a known adverse effect.

New warning: A precautionary statement is included about the potential risk of serious respiratory reactions. Healthcare professionals are advised to monitor patients and to discontinue the drug if respiratory symptoms develop.

Communicating to Healthcare Professionals

A label update alone is not enough. Regulators also ensure that those already prescribing Relvian are informed directly.

1. **Direct Healthcare Professional Communication (DHPC)**
   An advisory letter is sent to all relevant healthcare professionals, explaining the new safety finding, the updated warning, and the recommended steps to take when prescribing Relvian.
2. **Regulatory website update**
   The new information is published on the regulatory authority's website, making it publicly accessible to patients and professionals.
3. **Martin's doctor is informed**
   Martin's own doctor receives the advisory letter and reviews his prescribing practice accordingly. He also switches Martin to a safer alternative for his osteoarthritis.

OUTCOME FOR MARTIN

Martin is switched to a different osteoarthritis treatment: pain management with physical therapy support and an alternative analgesic. He remains stable with no further side effects.

CHAPTER 08

Why It Matters

Martin's story is not just about one patient and one medicine. It illustrates why pharmacovigilance exists and what it achieves for all of us.

★ The Ripple Effect of One Report

Martin's doctor filed a single report. That report contributed to a signal. The signal triggered an investigation. The investigation led to a label change. The label change informed thousands of prescribers. Future patients with Relvian prescriptions now carry a warning that could prompt their doctors to monitor more closely, stop the drug early, or choose a safer alternative.

One report. Real impact.

The Bigger Picture

Martin's case contributes to a better understanding of Relvian's safety profile. This is how pharmacovigilance works at scale: each report adds a small piece to a large and constantly evolving picture of medicine safety.

Protecting patients

By identifying rare risks early, pharmacovigilance prevents harm to patients who would otherwise be unaware of a danger.

Preserving trust

Transparent reporting and regulatory action maintain public confidence in medicines and the healthcare system.

Improving knowledge

Safety data collected after approval adds to the body of knowledge about how medicines behave in diverse real-world populations.

Informing decisions

Benefit-risk assessments depend on good safety data. Pharmacovigilance ensures those assessments remain current and evidence-based.

THE BOTTOM LINE

No medicine is without risk. What pharmacovigilance ensures is that the risks are known, communicated, and managed. It is one of the most important tools we have for keeping medicines as safe as possible for as many people as possible.

CHAPTER 09

Key Terms at a Glance

A quick reference for the pharmacovigilance terms used throughout this guide.

Adverse Event (AE)

Any undesirable medical occurrence in a patient who has been given a medicine. It does not necessarily mean the medicine caused it.

Adverse Drug Reaction (ADR)

A response to a medicine that is harmful and unintended, and where a causal relationship with the drug is at least reasonably possible.

Individual Case Safety Report (ICSR)

The standard document used to report a single patient's suspected adverse event to a regulatory authority or pharmaceutical company.

VigiBase

The WHO's global database of individual case safety reports. It holds over 30 million reports and is the largest pharmacovigilance database in the world.

Signal

Information suggesting a new or changed association between a medicine and an adverse event, requiring further investigation.

Pharmacovigilance

The science and activities relating to the detection, assessment, understanding, and prevention of adverse effects and other medicine-related problems.

MedDRA

Medical Dictionary for Regulatory Activities. The international standard terminology used to code adverse events so that reports can be compared across databases and countries.

Risk Minimisation

Measures taken to reduce the likelihood or severity of an adverse reaction, including label updates, educational materials, and healthcare professional communications.

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